AnchorMTM: Spatially Anchored Merge Tree Maps with Position Bounds

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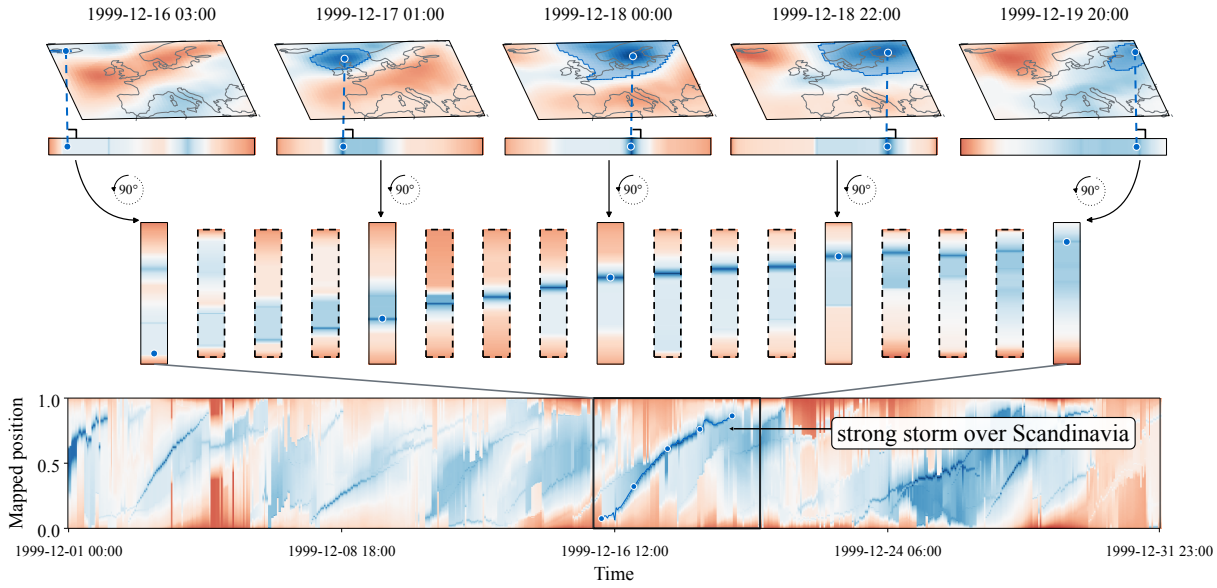


Figure 1: Qualitative Storms archive ($\varepsilon = 5\%$; preprocessing differs from Table 2). Spatial snapshots and profiles identify one matched extremum; blue outlines show leaf-arc supports. Rotated profiles and dashed intermediate samples connect to its trajectory in the full 744-frame map.

Abstract

Temporal merge tree maps show in one image how the topology of a time-varying scalar field evolves, but their display axis has no fixed spatial meaning: a feature's position cannot be read as its location, and a feature that moves need not move on the map. Anchoring the axis to a spatial direction conflicts with the merge tree, because an exact profile keeps every subtree contiguous while the projected locations of sibling subtrees can interleave. We ask how much displacement this conflict forces. We describe merge tree maps as one layout problem, in which TMTM and ST-MTM are exact-topology instances without a spatial reference, and prove three lower bounds on the displacement that exact topology, finite feature widths and a bounded merge error force on every instance. The bounds need no layout, so they measure any map and split its error into a part forced by the tree and a part chosen by its layout. AnchorMTM approaches the bounds with a single control, a merge-error budget. The bounds equal the exhaustive optimum on controlled trees and are attained on 606 of 614 planar frames of an earlier benchmark, and 31–95% of the order error of TMTM and ST-MTM is not forced by the tree. At the same zero merge error, AnchorMTM lowers the mean worst displacement by 73–96% on the planar ST-MTM benchmark and an hourly $PM_{2.5}$ sequence and by 63–67% on two volumes; a declared budget lowers it further, while ST-MTM's published neighborhood preservation remains higher.

Index Terms: Time-varying scalar fields, merge tree maps, topology preservation, spatial fidelity, lower bounds, constrained layout.

1 Introduction

An analyst who follows storms across an ocean or fire activity across a region reads a temporal overview for three things at once: which features exist and how they nest, where they are, and how they move. Animation keeps the spatial view of each instant but makes distant times hard to compare; static overviews give one display axis to time and compress space into the other [2, 47, 53]. What that second axis means decides which of these questions the overview can answer. Figure 1 links the spatial supports of one feature to such a map.

Merge tree maps answer the first question well. Temporal Merge Tree Maps (TMTM) draw the merge tree of each frame as a 1D scalar profile and align the profiles along time, so minima show extrema and the barriers between them show merge levels [31]. Spatiotemporal Merge Tree Maps (ST-MTM) add relative spatial distances and temporal position constraints to the layout [10]. Neither gives its display axis a fixed spatial meaning: TMTM orders leaves for the tree and for temporal alignment, and pairwise distances fix relative positions but no origin or direction. A feature's position therefore cannot be read as its longitude or its distance along a flow, and a feature that moves east need not move on the map. Hovmöller diagrams take the opposite side: they keep one physical coordinate and average the field over the others, so nearby extrema blend and the hierarchy disappears [26].

Anchoring the axis means projecting every extremum onto one direction and placing it near its projection. The merge tree resists.

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An exact profile keeps every subtree in one contiguous block, but the projections of sibling subtrees can interleave. In Figure 2, four minima project in the order A, C, D, B, which separates A and B by the C, D subtree, so some features must leave their references or some merge levels must change; finite feature widths add packing constraints. ST-MTM observes that a one-dimensional layout cannot preserve all spatial distances, so some distortion is an expected consequence of the projection [10]. How much of it the hierarchy forces, where it arises and how much merge error buys it back have not been quantified, yet they decide whether a spatial axis is worth having.

We answer these questions in three steps. First, we write merge tree maps as one layout problem: a leaf order, intervals and anchors that minimize a weighted sum of positional, geometric and temporal terms under a merge-error allowance. TMTM and ST-MTM are its exact-topology instances without a spatial reference term, and projections and space-filling curves sit at its other end. Second, we bound what every instance must pay. Three lower bounds give the displacement that exact topology, finite widths and a bounded merge error force; they depend only on the tree, the projected locations and the widths, so they measure any map, name the merges that cause the displacement and the merge error that removes it, and split the error of a map into a part forced by the tree and a part chosen by its layout. Third, AnchorMTM, *Spatially Anchored Merge Tree Maps*, is the instance with a reference term and one control, a merge-error budget ε, that approaches the bounds. Exact Mode ($\varepsilon = 0$) keeps the tree, as TMTM and ST-MTM do; Budgeted Mode trades a declared and verified merge error for position.

Our main contributions are as follows:

- **A common formulation with position bounds.** TMTM and ST-MTM are exact-topology instances of one layout problem; three layout-independent lower bounds give the displacement that exact topology, finite widths and a bounded merge error force on every instance (Sections 3 and 4).
- **AnchorMTM.** A spatially anchored construction with one control, a merge-error budget ε, that approaches the bounds (Section 5).
- **An evaluation on planar and volumetric data.** The bounds are tight; 31–95% of the order error of TMTM and ST-MTM is not forced by the tree; at the same zero merge error, AnchorMTM lowers the mean worst displacement by 73–96% on the four ST-MTM benchmark inputs and hourly $PM_{2.5}$ and by 63–67% on two volumes (Section 6).

2 Related Work

2.1 Temporal Overviews of Spatial Data

Animation keeps the spatial view of each instant, whereas static overviews expose distant times at once; which serves better depends on the task [47, 53], and generalized space–time cubes describe the operations that lead from one to the other [2]. A static overview must compress space into one display axis. Hovmöller diagrams average the field across a chosen direction [26]; dense pixel displays, geographic orderings and space-filling curves linearize the domain [9, 21, 46], and volume lines and data-driven curves do so for volumes [59, 66]. For moving entities, one-dimensional projections and visual summaries such as MotionRugs, stable summaries and MoReVis order objects by position or stability [6, 19, 54, 62]. Every such linearization distorts some spatial relation [1, 41]. These displays keep a spatial coordinate but no feature hierarchy; we keep the merge tree and quantify what the coordinate then costs.

2.2 Topology of Time-Varying Scalar Fields

Merge and contour trees summarize how level sets connect, and persistence-based simplification controls their size [14, 22, 52]. Features are tracked over time by overlap or attribute similarity [48, 49] and drawn as tracking graphs or nested tracking graphs [37, 38, 60]; time-varying merge trees are computed and compared with Wasserstein and geometry-aware interleaving distances [42, 44, 63, 65], and such pipelines follow cyclones and atmospheric rivers [32, 64]. Labelled interleaving distances [20, 39, 40] and ordered merge trees [5] underlie the merge error we bound. Tracking graphs show how features appear, split and merge but not where they are; merge tree maps, discussed next, also show the values and nesting of every frame.

2.3 Drawing Hierarchies in One Dimension

Topological landscapes and their one-dimensional profiles draw a merge tree as terrain or a curve whose minima are the extrema and whose barriers are the merge levels [43, 57], with geometry-preserving and temporally coherent variants [4, 24]; contour trees also have planar drawings and joint layouts [23, 36]. Temporal treemaps place evolving hierarchies along time [30]. TMTM aligns the profiles of successive merge trees into one static map [31], and ST-MTM adds relative spatial distances and temporal stability [10]. Storyline layouts face a related problem for groups of entities, trading crossings and wiggles against group contiguity [35, 51]. TMTM and ST-MTM are the only methods that produce the output we study, a static map of merge tree profiles over time, so they are our direct baselines; Table 1 places them with tree-only orderings and space-only linearizations in one formulation, and Section 6 measures all of them.

2.4 Hierarchies against an External Order

The conflict we bound recurs wherever a tree must respect an outside order. Optimal leaf ordering of dendrograms and clustered heat maps chooses a tree-consistent order for adjacent similarity [3, 15], and reordering toward a target finds the closest tree-consistent permutation [7]. Spatially ordered treemaps and data-spatial grid maps trade hierarchy or data layout against geographic position [55, 61]. Geophylogenies connect leaf orders to map locations [28], and tanglegrams minimize crossings or displacement between two trees [18, 27]. Separation constraints keep laid-out objects disjoint [13, 34], ultrametric fitting under the maximum norm relates to our optimal barriers [17], and merge distortion and topology-preserving projections measure how much hierarchy a mapping keeps [12, 16]. These works optimize crossings, sums or stress for a given instance; we give lower bounds, in closed form for points, on the worst displacement that every tree-consistent layout must accept, together with the merge error that removes it.

3 Spatially Anchored Merge Tree Maps

We first describe merge tree maps in one formulation that covers the existing ones, then give the axis a spatial reference and show where it conflicts with the tree.

3.1 Merge Trees and One-Dimensional Profiles

For a scalar field f_t on a 2D or 3D domain, the merge tree T_t records how sublevel-set components appear at minima and join at saddles as the threshold increases; negating the field gives the split tree [22, 52]. Persistence simplification precedes layout, and all statements refer to the simplified tree [14]. Each leaf i is an extremum at location p_i^t; its support S_i^t is the set of samples on its leaf arc, with measure m_i^t. Two leaves merge at the level $V_{ij}^t = h_t(\mathrm{lca}(i, j))$ of their lowest common ancestor, and V_{ii}^t is the extremum value; equal-level merges form one multifurcation.

A one-dimensional profile draws the leaves as minima in a chosen leaf order, each inside an interval sized by its support, and fills the gaps from the tree, so that the largest profile value between two minima shows their merge level [31, 43, 57]. The profile shows the tree exactly only if every subtree occupies one contiguous block of leaves. Stacking the profiles along time gives a temporal merge tree

map; leaves of consecutive frames are matched by support overlap with a one-to-one assignment, and $\mu_t(i)$ denotes the matched predecessor of leaf i [10].

3.2 A Common Formulation

A map frame consists of a leaf order π, intervals with centers z_i and widths W_i, display anchors x_i for the extrema, and a profile ψ. The layout set $\mathscr{L}(\pi;W)$ contains the placements whose intervals follow π on the canvas $[0,L]$, are disjoint with gap at least g, and hold each anchor within $\rho W_i/2$ of its center. Two errors describe a frame. The *merge error* compares displayed and true merge levels of fixed labels,

$$d_{\mathrm{top}}(\psi,T) = \max_{i,j} \left| V_{ij}^{\psi} - V_{ij} \right|, \tag{1}$$

where V_{ij}^{ψ} is the largest profile value between the anchors of i and j. It is zero exactly when the profile shows the tree, and for complete labels it equals the labelled interleaving distance, which bounds the interleaving and bottleneck distances [20,39,40]. The *positional error* compares anchors with spatial targets q_i (Section 3.3); its worst value $\|x-q\|_\infty$ is reported per frame. Two order-level versions isolate the role of the leaf order:

$$\begin{aligned} \mathrm{NROE}(\pi) &= \frac{1}{2L} \max_{j \le k} \left[q_{\pi(j)} - q_{\pi(k)} \right]^+, \\ \tau(\pi) &= \min_{(x,z) \in \mathscr{L}(\pi;W)} \|x-q\|_\infty. \end{aligned} \tag{2}$$

NROE is the smallest worst displacement of points placed in order π, relative to L; τ adds the intervals and is computed exactly for a fixed order.

Each map we consider solves, frame by frame, an instance of

$$\begin{aligned} \min_{\pi \in \Pi,\ (x,z) \in \mathscr{L}(\pi;W)} \ & \lambda_R E_R + \lambda_G E_G + \lambda_M E_M + \lambda_C E_C \\ \text{s.t. } & d_{\mathrm{top}}(\psi,T) \le \eta. \end{aligned} \tag{3}$$

Here E_R is the mean squared distance of anchors to their references, E_G a stress between anchor separations and support-centroid distances, E_M the mean squared distance to a temporal target $\hat{x}_i$ derived from the matched previous anchor, and E_C the offset of anchors from interval centers. A method is a choice of order rule Π, weights λ, temporal target $\hat{x}$, width semantics W and allowance η (Table 1). TMTM and ST-MTM keep the tree ($\eta = 0$) and have no reference term ($\lambda_R = 0$): TMTM greedily orders children by subtree overlap between frames and packs intervals by sample count [31]; ST-MTM picks a temporally concordant or optimal leaf order within the tree and fits relative distances [10]. Their axes therefore encode tree traversal or relative distance rather than a location. Projections and space-filling curves sit at the other end: they follow space and ignore the tree [19]. AnchorMTM adds the reference term and a declared allowance η, controlled by one number ε (Section 5). The correspondence concerns the problem each method solves; solvers differ in scaling and other details, and Section 6 evaluates implementations of the published methods, not our solver with their weights.

3.3 Spatial Reference

A reference gives the axis a meaning: $q_i^t = \langle a, p_i^t \rangle - o$ for one unit direction a and one origin o, where o and the extent L come from the projected corners of the domain. By default, a is the direction of largest matched extremum motion,

$$a^* \in \arg\max_{\|a\|=1} \ a^\top \Big(\sum_{t,i} \Delta p_i^t (\Delta p_i^t)^\top \Big) a, \tag{4}$$

with $\Delta p_i^t = p_i^t - p_{\mu_t(i)}^{t-1}$ over valid matches, signed so that the total projected motion is nonnegative; a direction chosen by the analyst can replace it. Direction, origin and extent are fixed for the whole sequence and all budgets, with no per-frame recentering, so a feature that moves along a moves along the axis. Matching by support overlap can miss fast features, which then contribute no motion sample.

Table 1: Maps of time-varying scalar fields as instances of Eq. 3. Order rule: how leaves are ordered; placement: nonzero terms and temporal target; η: allowed merge error relative to the scalar range (tolerance ε_0 omitted); axis: what a position encodes.

Method	Order rule	Placement	η	Axis
TMTM [31]	greedy child order by subtree overlap between frames	packed by sample count; no anchor optimization	0	tree traversal
ST-MTM [10]	concordant order unless OLO is cheaper by a ratio	λ_G (stress), λ_M on matched anchors, then intervals	0	relative distance
OLO [3]	tree-constrained OLO	ranks only	0	order
Projection, Hilbert, Morton [19]	by reference or curve	at reference or curve rank	none	location or curve rank
AnchorMTM	least τ within the budget	λ_R dominant, $\hat{x}$ follows projected motion	ε	location

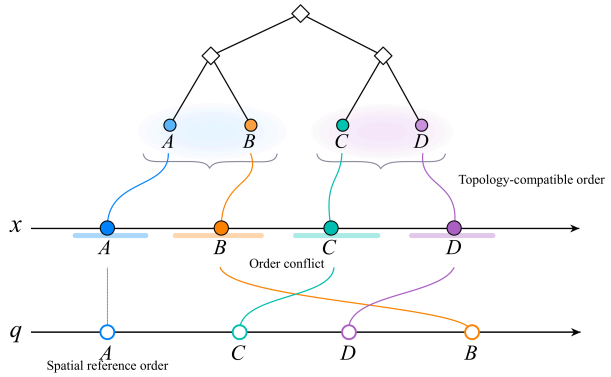


Figure 2: Reference–topology conflict. The tree requires contiguous A,B and C,D blocks, but the references occur as A,C,D,B. Any exact profile must displace some anchors (colored links); the bounds of Section 4 give the least such displacement. Widths and positions are schematic.

3.4 Where Topology and Space Conflict

An exact profile keeps every subtree contiguous, but the references of sibling subtrees can interleave. In Fig. 2 the references occur in the order A,C,D,B, which separates A and B by the C,D subtree, so no exact profile places all four anchors at their references; finite widths add packing constraints. How large must the displacement be, where does it arise, and how much merge error removes it? The answer depends only on T, q, W and η, so it holds for every instance of Eq. 3 at once.

4 The Positional Price of Keeping the Tree

The three bounds below involve the tree, the references, the widths and the allowance, but neither the order rule nor the weights, so they hold for every row of Table 1. The first concerns exact topology, the second adds widths, and the third allows a merge error. Full proofs are in Supplementary Material; we drop t within a frame, write $[v]^+ = \max(v,0)$ and take empty maxima as zero.

4.1 Exact Topology

On the reduced tree, where equal-level merges form one node, exact topology makes every subtree contiguous. Let $[m_X, M_X]$ be the reference range of the leaves of subtree X.

Theorem 1 (Exact-Topology Position Bound)**.** *Every profile with $d_{\text{top}} = 0$ has $\text{NROE}(\pi) \geq \nu^*(T,q)$, where*

$$\nu^*(T,q) = \frac{1}{2L} \max_{v} \max_{\substack{X,Y \in \text{ch}(v) \\ X \neq Y}} \big[\min(M_X - m_Y, M_Y - m_X)\big]^+. \quad (5)$$

Ordering children recursively by the midpoints of their reference ranges attains ν^* in $O(n \log n)$ time.

Proof sketch. Two sibling blocks must take one of two orientations, with inversion costs $[M_X - m_Y]^+/2$ and $[M_Y - m_X]^+/2$; midpoint order takes the smaller, and every inverted pair of leaves is separated by sibling blocks at its lowest common ancestor. □

The maximizing node and sibling pair say where the price arises, at the merge of two subtrees whose reference ranges overlap, and the value says how large it is: half the overlap left in the cheaper orientation.

Forced and chosen error. Because midpoint ordering attains ν^* on the same tree, the order error of any exact-topology map splits into $\nu^*(T,q)$, which the tree forces, and $\text{NROE}(\pi) - \nu^*(T,q)$, which its order rule chooses and which can be removed without changing the tree. NROE does not involve widths, so the split applies equally to TMTM and ST-MTM, whose interval sizes follow different rules.

4.2 Finite Widths

Intervals need room: a block of leaves occupies at least the sum of its widths and gaps. For disjoint leaf sets X and Y, let $\omega(X,Y)$ be the least worst displacement that placing all of X before all of Y, or the reverse, forces on their anchors, whichever is smaller, and let E_∂ be the cost of fitting all intervals between the canvas ends. Both follow from sorted reference endpoints and cumulative widths (Supplementary Material).

Theorem 2 (Width-Aware Position Bound)**.** *Let $\mathscr{P}_0$ contain every pair of descendant-leaf sets of two siblings. Every exact-topology interval layout satisfies*

$$\begin{aligned} \|x - q\|_\infty \geq \tau(\pi) &\geq \mathscr{B}_w(T,q,W) \\ &= \max\Big\{E_\partial, \max_{(X,Y)\in\mathscr{P}_0} \omega(X,Y)\Big\}. \end{aligned} \quad (6)$$

With zero widths and gaps, $\mathscr{B}_w = L\nu^*(T,q)$.

Proof sketch. The first and last anchors of a block must enclose its intervals and gaps up to the anchor slack, which gives a shortfall against the reference separation; exact topology fixes one orientation of every sibling pair, and the canvas ends give the one-sided costs. □

$\mathscr{B}_w$ bounds every exact-topology order and need not be attained; $\tau(\pi)$ is the attained minimum for a given order.

4.3 A Bounded Merge Error

A merge allowance can release a block but still limits interleaving. If leaf k lies between two descendants i, j of node u, its displayed merge with i cannot exceed theirs, so $V_{ik} \leq h(u) + 2\eta$. Leaf k can therefore enter the block of u only if it descends from $A_\eta(u)$, the highest ancestor of u whose level is within 2η of $h(u)$. Let $\mathscr{P}_\eta$ contain the leaf-set pairs this rule keeps apart: $(\mathscr{D}(u), \{k\})$ for leaves k outside $\mathscr{D}(A_\eta(u))$, and $(\mathscr{D}(u), \mathscr{D}(w))$ for internal nodes u, w with $\mathscr{D}(w)$ disjoint from $\mathscr{D}(A_\eta(u))$ and $\mathscr{D}(u)$ disjoint from $\mathscr{D}(A_\eta(w))$, where $\mathscr{D}(\cdot)$ denotes descendant leaves.

Theorem 3 (Bounded-Topology Position Bound)**.** *In every profile with $d_{\text{top}} \leq \eta$, each pair in $\mathscr{P}_\eta$ lies wholly in one orientation. Hence*

$$\begin{aligned} \text{NROE}(\pi) \geq C_\eta &= \max_{(X,Y)\in\mathscr{P}_\eta} \frac{[\min(M_X - m_Y, M_Y - m_X)]^+}{2L}, \\ \tau(\pi) &\geq \max\Big\{E_\partial, \max_{(X,Y)\in\mathscr{P}_\eta} \omega(X,Y)\Big\}. \end{aligned} \quad (7)$$

The theorem also says what releases a conflict: a subtree may interleave with its sibling once 2η reaches the gap between its own merge level and its parent's, so conflicts at deep merges stay expensive. Flattening every parent gap of at most 2η gives a second valid bound; neither dominates, and we report the larger. Read in reverse, the theorem prices the other end of Table 1: a map whose order error is below C_η, such as a projection with zero order error, must show a merge error above η.

5 AnchorMTM: Approaching the Bounds

AnchorMTM is the instance of Eq. 3 with a reference term and the allowance $\eta = (\varepsilon_0 + \varepsilon)\Delta_F$, where Δ_F is the scalar range of the sequence, $\varepsilon_0 = 10^{-7}$ a fixed numerical tolerance, and $\varepsilon \geq 0$ the only control. *Exact Mode* ($\varepsilon = 0$) keeps the tree up to ties within $\varepsilon_0\Delta_F$; *Budgeted Mode* ($\varepsilon > 0$) trades a declared merge error for position. Since Eq. 3 couples a discrete order with continuous placement, AnchorMTM solves it in two stages that follow the bounds (Fig. 3, panel 2): it chooses the order by τ, the quantity the bounds constrain, and then places anchors within a cap on τ. Intervals have widths $W_i = s\,m_i$ with one fixed scale s; all other settings form one fixed configuration, listed in Supplementary Material.

5.1 Leaf Order within the Budget

For each frame, the order is chosen lexicographically,

$$\pi^\varepsilon \in \arg\min_{\pi\in\mathscr{C}} \big(\tau(\pi), \text{NROE}(\pi)\big) \quad \text{s.t.} \quad \delta^*(\pi) \leq \eta, \quad (8)$$

over a finite candidate pool $\mathscr{C}$: the exact tree ordered by midpoints (which attains ν^*), threshold and conflict-driven flattenings of the tree, and the orders visited while width-aware block moves refine the best of them. Every candidate is evaluated against the original merge matrix. The pool is heuristic, so the selected order is not a certified optimum over all permutations; Section 6 measures how often it attains the bounds. Admission uses the smallest merge error that an order can show.

Proposition 1 (Optimal Barriers)**.** *For a fixed order, let $C_k = \min_{i\leq k<j} V_{\pi(i)\pi(j)}$ be the lowest merge across gap k and F_k the larger of its two adjacent extremum values. The smallest attainable merge error is*

$$\begin{aligned} \delta^*(\pi) = \max\Big\{0,\ &\max_k (F_k - C_k), \\ &\tfrac{1}{2}\max_{i<j}\big(V_{\pi(i)\pi(j)} - \max_{i\leq k<j} C_k\big)\Big\}, \end{aligned} \quad (9)$$

attained by the barriers $b_k = C_k + \delta^*(\pi)$.

An order therefore meets the budget exactly when $\delta^*(\pi) \leq \eta$, which makes admission a closed-form test.

5.2 Anchor Placement

For the selected order, AnchorMTM solves Eq. 3 with default weights $(\lambda_R, \lambda_G, \lambda_M, \lambda_C) = (4, 1, 0.5, 0.1)$, the temporal target $\hat{x}_i^t = x_{\mu_t(i)}^{t-1} + (q_i^t - q_{\mu_t(i)}^{t-1})$, which follows projected motion and keeps $x - q$ continuous, and the additional cap $\|x - q\|_\infty \leq \tau(\pi) + \beta_0$ with a fixed slack β_0. The cap is feasible because τ is attained, and the

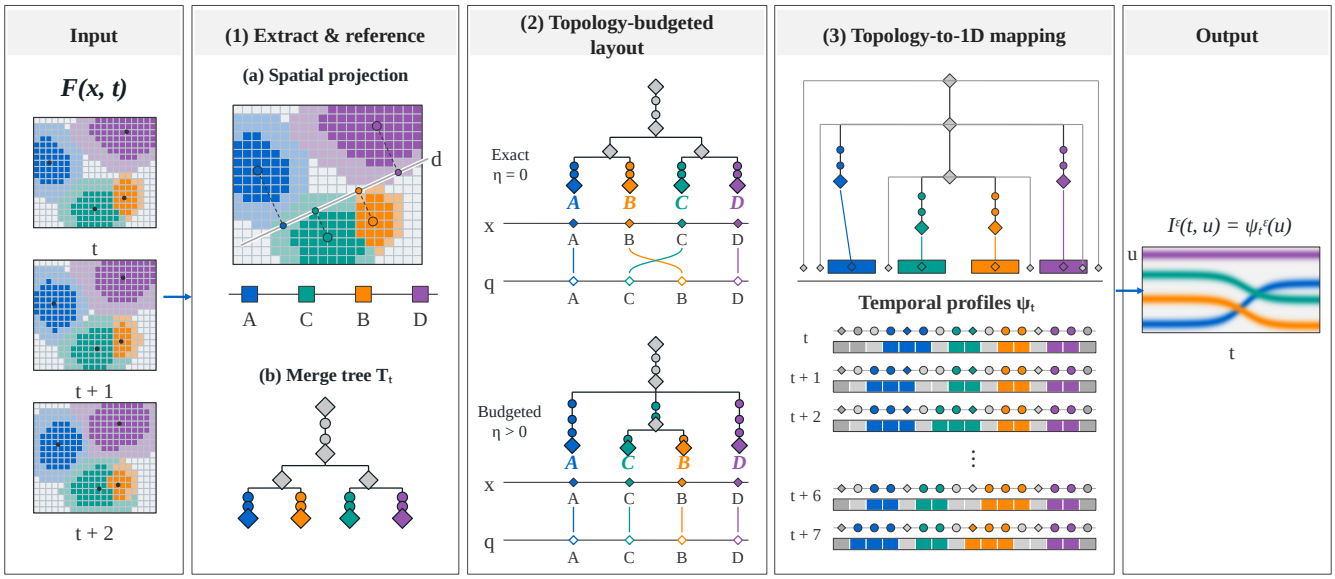


Figure 3: AnchorMTM construction (schematic). (1) Extract and reference: merge trees of each frame, leaf supports matched across frames, and one fixed direction d (the a^* of Eq. 4) that gives every extremum its reference position q. (2) Topology-budgeted layout: search leaf orders admitted by the budget η, then place anchors x under a displacement cap; with $\eta = 0$ the order keeps the subtree blocks and anchors leave their references, while $\eta > 0$ admits an order that follows them. (3) Topology-to-1D mapping: fill intervals and barriers into each profile ψ_t and stack the profiles over time into the map $I^\varepsilon(t,u) = \psi_t^\varepsilon(u)$.

problem is a convex quadratic program for a fixed order. The program cannot replace the selected order, so the cap transfers the order's positional guarantee to the anchors. Leaves without a matched predecessor have no temporal term.

5.3 Profile Reconstruction and Checks

Profiles reuse the leaf-arc sampling and the filling along paths to the lowest common ancestor of merge tree maps [10, 31]; extrema stay at their anchors, and each gap is then clipped or raised to its barrier b_k, which relaxed orders require. All frames share the direction, origin, extent and pixel scale, with no per-frame recentering. Because raster rounding and strict-extremum clearance can change the displayed merge levels, every rendered frame is checked against the original merge matrix, and records keep the requested budget, τ, δ^* and the measured errors. Capacity, input and solver failures are reported rather than repaired by rescaling.

6 Evaluation

We ask four questions. **Q1:** are the bounds tight? **Q2:** how much of the positional error of existing maps does the tree force? **Q3:** at the same topological guarantee, does AnchorMTM place features closer to their references, and does it keep its budget? **Q4:** what does a budget buy, frame by frame? Section 6.6 then reads the maps.

6.1 Data and Protocol

Table 3 lists two groups of sequences. The *bound suite* comes with the saved trees of an earlier benchmark, prepared with one extractor, matching rule and reference estimator and with one default configuration; on it we evaluate the bounds, all leaf orders of small frames and every compared map frame by frame (Q1, Q2). The *layout suite* reconstructs the four planar inputs of the ST-MTM benchmark [10], so that AnchorMTM is compared on the data of the closest prior work, and adds an hourly $PM_{2.5}$ sequence outside that benchmark and the two TMTM volumes [31, 58] (Q3, Q4). The hourly $PM_{2.5}$ sequence covers January 2017 over northern China (110–120°E, 32–41°N) with 744 reanalysis fields [29] interpolated to 160×144; maxima are extracted as join trees of the negated field, with persistence 5% of each frame's range. Ring, Storms, Wildfires and the volumes occur in both suites with different preprocessing (Supplementary Material); the two $PM_{2.5}$ sequences are different data, daily in the bound suite and hourly in the layout suite. Author-processed arrays are unavailable, so the layout suite is a disclosed reconstruction, with the ST-MTM persistence numbers read as per-frame range fractions (0.1, 0.1, 15 and 10.5%), not an identical-input reproduction.

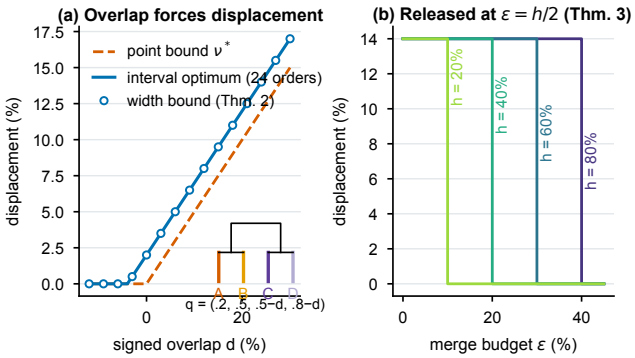


Figure 4: Controlled four-leaf trees (inset: sibling pairs and references). (a) Increasing sibling-range overlap d forces displacement; the width bound of Theorem 2 (circles) equals the optimum over all 24 leaf orders. (b) At $d = 24\%$, the 14% displacement disappears exactly when the merge budget reaches half the parent gap h, for all four gaps, as Theorem 3 predicts. Percentages refer to a unit canvas or a unit scalar range.

Measures. NMLE is the attained merge error d_{top} (Eq. 1) of the rendered profile divided by the scalar range, and NROE the or-

Table 2: Five-dataset comparison. Bold/underline mark the best/second distinct local measurements per dataset; ties share emphasis. Curves use fixed default orientations.

Dataset	Hilbert			Morton			TMTM			ST-MTM			AnchorMTM (ours)			
	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	SNS↓	TW↑	NPE↓	Mode	SNS↓	TW↑	NPE↓
Gaussian	0.1117	0.6719	0.7557	<u>0.0393</u>	<u>0.8333</u>	0.7359	0.2500†	0.7700†	<u>0.1118</u>	0.0600†	1.0000†	0.1154	Exact	**0.0353**	**0.9323**	**0.0089**
													Balanced	**0.0353**	**0.9323**	**0.0089**
Ring	**0.1252**	**0.9048**	0.5941	0.1489	0.7619	0.6033	0.2800†	0.6800†	0.1756	0.2100†	0.8800†	0.2463	Exact	0.1485	<u>0.8571</u>	<u>0.0471</u>
													Balanced	<u>0.1334</u>	<u>0.8571</u>	**0.0395**
Wildfires	0.1883	0.8303	0.3641	0.1304	0.8758	0.1560	0.4400†	0.8800†	0.2280	0.2400†	0.9100†	0.3180	Exact	**0.0426**	**0.9136**	<u>0.0131</u>
													Balanced	<u>0.0447</u>	<u>0.8989</u>	**0.0116**
Storms	0.1290	0.7403	0.3705	0.1180	0.7251	0.3468	0.3400†	0.7100†	0.5753	0.2100†	0.9400†	0.4579	Exact	<u>0.0790</u>	<u>0.8463</u>	0.0399
													Balanced	**0.0743**	**0.8766**	**0.0140**
$PM_{2.5}$	0.2386	**0.8887**	0.8090	<u>0.1760</u>	<u>0.8277</u>	0.8823	0.3018	0.7781	0.5241	0.1897	0.8224	0.5577	Exact	0.1964	0.7884	<u>0.1390</u>
													5%	**0.1636**	0.7657	**0.0468**

†Grey: published TMTM/ST-MTM Temporal SNS/TW on the four benchmark inputs (ST-MTM Table 1, authors' inputs), shown for reference and excluded from emphasis; all other entries, including every $PM_{2.5}$ value, are local measurements on the same inputs. Author inputs and coverage are unconfirmed. Local TW coverage (Gaussian/Ring/Wildfires/Storms/$PM_{2.5}$) is 64/64, 1/40, 4/32, 11/744, 744/744; ineligible frames are omitted. Published TW has two-decimal precision. Mode applies only to AnchorMTM; $PM_{2.5}$ uses a fixed 5% budget, not a selected Balanced budget.

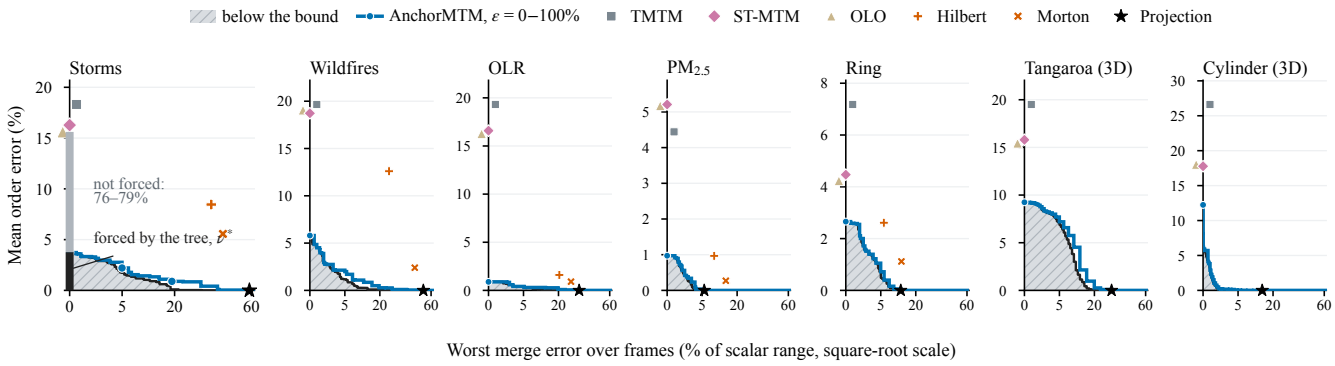


Figure 5: Position–topology plane of the bound suite, five planar and two volumetric sequences. Each map is one point: its worst merge error over frames (horizontal, square-root scale) and its mean order error (vertical). No map can enter the grey region below the bound (Theorem 3 and the flattened bound; Theorem 1 at zero merge error). TMTM, ST-MTM and OLO keep the tree and sit at zero merge error far above the bound; on Storms 76–79% of their order error is not forced by the tree. Space-filling curves and the projection follow space and break the tree. AnchorMTM (blue, ε from 0 to 100%) runs along the edge; on Storms, OLR, daily $PM_{2.5}$ and Ring, enumerating all leaf orders shows that it is the exact optimum at every tested budget except one Storms frame. TMTM, ST-MTM and OLO are shown in the mirror that agrees better with the reference.

der error of Eq. 2. NPE is the worst displacement $\max_i |x_i^t - q_i^t|$ of a frame divided by the physical extent and averaged over frames. SNS is the scale-normalized stress between support-centroid and anchor distances [10, 50], and TW measures neighborhoods [56] with $k = (1,3,5,3)$, using only frames with $0 < k < n_t/2$ (64/64, 1/40, 4/32 and 11/744), and $k = 1$ on all 744 $PM_{2.5}$ frames. On the four benchmark inputs, the SNS and TW of TMTM and ST-MTM in Table 2 are quoted from ST-MTM (Temporal mode, authors' inputs) and are not ranked against local values. All other baseline values are measured on our implementations of the published methods with the same trees, identities and references, with ST-MTM's Storms parameters on $PM_{2.5}$. Each baseline map in Table 2 is reflected once for the whole sequence if its anchors and references correlate negatively on the first frame where both vary; Figs. 5 and 6 use the global mirror that agrees better with the reference. Hilbert and Morton curves keep a fixed grid orientation [19]. The volumetric table uses the $+x$ axis chosen by an offline displacement rule and stored, tuned configurations (Supplementary Table S4), whereas Figs. 6 and 7 use the default configuration of the bound suite.

Modes. Exact is Exact Mode ($\varepsilon = 0$). Balanced is Budgeted Mode with a positive budget selected from 1–10% on the first chronological quarter by $J = 0.5\,\mathrm{NPE}_{norm} + 0.5\,\mathrm{SNS}_{norm}$ over candidates within their budget, prefers smaller budgets within 0.01 of the minimum, and is frozen before the full runs: 1, 1, 5 and 10% (Supplementary Fig. S4). Earlier outputs had been observed, so this is not a blind test. $PM_{2.5}$ was added later with a fixed $\varepsilon = 5\%$ that this rule did not select, and it is reported under its budget rather than as Balanced.

6.2 Q1: Are the Bounds Tight?

Controlled trees. We fix two sibling pairs $\{A,B\}$ and $\{C,D\}$ that merge at 0.4 below a root at 1, with leaf values 0 and references $(0.2, 0.5, 0.5-d, 0.8-d)$ on a unit canvas, and vary the signed overlap d from -0.12 to 0.30. Across 43 overlaps and four widths, we compare the bounds with all 24 leaf orders, each placed by an independent minimax LP. Topology alone forces exactly $v^* = \max(0,d)/2$. With widths 0.06, gap 0.01 and $\rho = 0.5$, the width bound, our search and the exhaustive optimum coincide at $\tau = \max(0,(d+0.04)/2)$ (Fig. 4a): 12% for points and 14% for intervals at $d = 0.24$. A merge budget removes the conflict exactly where Theorem 3 admits interleaving, at $2\varepsilon = h$ for all four parent gaps h (Fig. 4b).

Real trees. In Exact Mode, the interval cost τ of the selected order equals $\mathscr{B}_w$ on 606 of the 614 planar frames of the bound suite, so on these frames no exact-topology order has a smaller worst displacement; at $\varepsilon = 10\%$ the bounded-topology bound is met on 596. On Storms, OLR, daily $PM_{2.5}$ and Ring every frame has at most

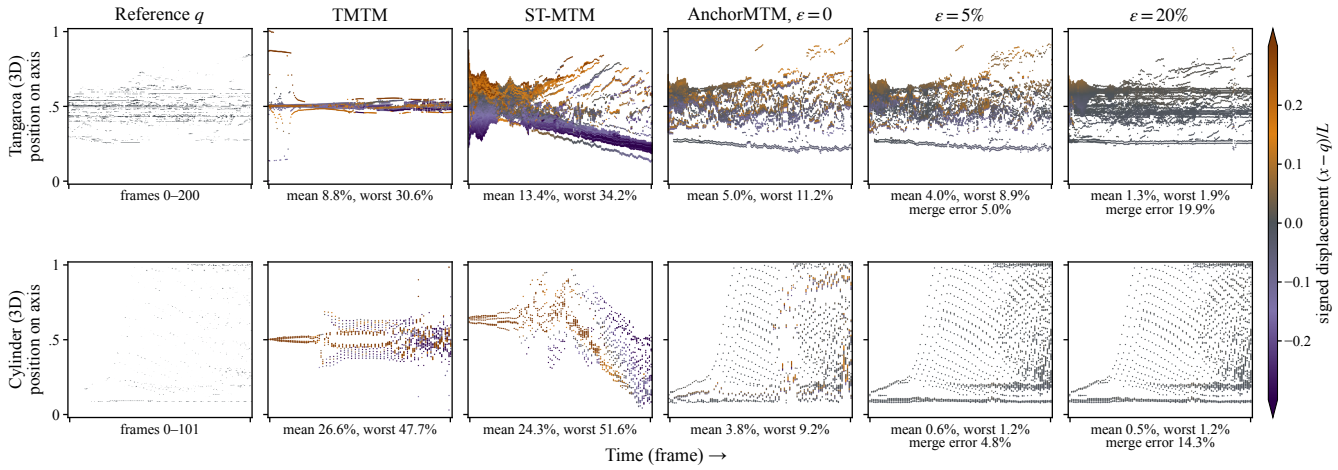


Figure 6: Where each method draws the features of the two volumes. Each dot is one feature in one frame at its position on the axis (fraction of the canvas), coloured by its signed displacement from its reference; grey dots lie on their references. The first column shows the references, i.e., what a map read as location should show. All methods use the same trees and features; TMTM and ST-MTM are implementations of the published methods, each in the mirror that agrees better with the reference. AnchorMTM uses the default configuration of the bound suite with the matched-motion direction (Tangaroa) or the streamwise x axis (Cylinder). Below each panel: mean and mean worst displacement over frames and, for a positive budget, the attained merge error.

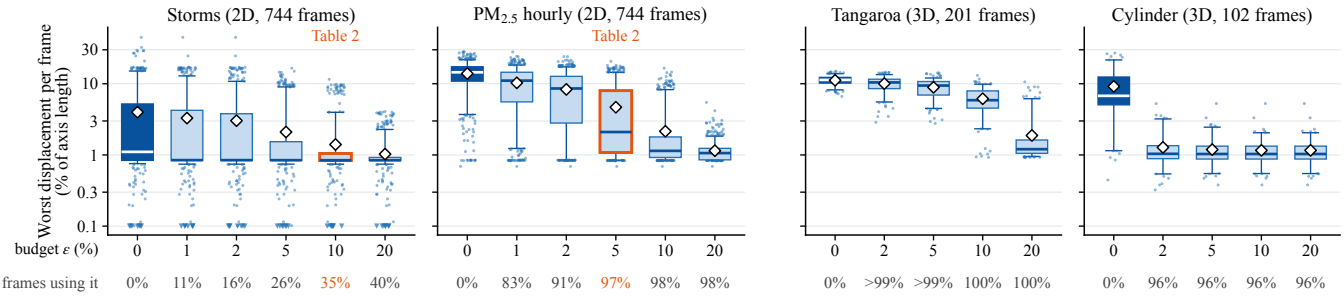


Figure 7: Per-frame worst displacement of AnchorMTM over the merge budget ε on complete sequences (log scale): two planar inputs with the configuration of Table 2 and the two volumes with the default configuration. Boxes: quartiles; whiskers: 5th–95th percentiles; dots: other frames (triangles: seven Storms frames below 0.1%); diamonds: means, the NPE of the tables; dark: Exact; orange: the budget of Table 2. The bottom row gives the share of frames whose attained merge error is positive; no frame exceeds its budget. On Storms a budget shortens the upper tail; on $PM_{2.5}$ and in the volumes almost every frame uses it and the whole distribution moves. Wildfires, Ring and per-frame merge errors: Supplementary Figs. S1–S3.

eight leaves, and enumerating all leaf orders shows that the AnchorMTM order reaches the least order error at every tested budget, except one Storms frame at $\varepsilon = 31.6\%$ (1.9 points). Volumes are harder: the width bound is met on 81 of 102 Cylinder and 27 of 201 Tangaroa frames, and the remaining gap may lie in the bound or in the heuristic search.

6.3 Q2: How Much of the Error Does the Tree Force?

Theorem 1 needs no layout, so it measures every exact-topology map, and midpoint ordering attains v^* on the same tree. On the seven bound sequences, 53–95% of the order error of TMTM and 31–95% of that of ST-MTM lies above v^* and is therefore chosen by their order rules (Supplementary Table S2); OLO, which only optimizes adjacency within the tree, behaves alike (32–95%). Figure 5 places every map in the position–topology plane. Maps that keep the tree sit at zero merge error far above the bound; the projection reaches zero order error only with merge errors of 5.6–59%, and the space-filling curves also break the tree (8–40%); AnchorMTM runs along the edge between them. The edge shows what a budget releases: where conflicting subtrees merge just below their parent, 1% halves the forced order error on Wildfires (5.80% to 2.77%) and almost removes it on Cylinder (12.24% to 0.23%), whereas the deep conflicts of Tangaroa keep it above 7% up to $\varepsilon = 5\%$. The forced cost also depends on the direction and can be computed before any map is drawn: over 400 directions it spans 4.70–10.14% on Tangaroa and 5.49–28.15% on Cylinder (Supplementary Fig. S6).

6.4 Q3: Closer at the Same Guarantee?

Planar inputs. At the same zero merge error, Exact lowers NPE relative to TMTM and ST-MTM by 73–96% on all five inputs of Table 2 (73.5 and 75.1% on $PM_{2.5}$), and it is also below the Hilbert and Morton curves, which break the tree. Balanced, with a budget of at most 10%, lowers NPE relative to Exact by a further 16.2% on Ring, 11.6% on Wildfires and 64.9% on Storms; Gaussian is unchanged, and the fixed 5% budget lowers it by 66.3% on $PM_{2.5}$. This supports alignment to a chosen reference, not general quality. ST-MTM reports higher TW on all four benchmark inputs, and we do not rank its published SNS and TW, which come from the authors' inputs and coverage, against local values. On $PM_{2.5}$, where

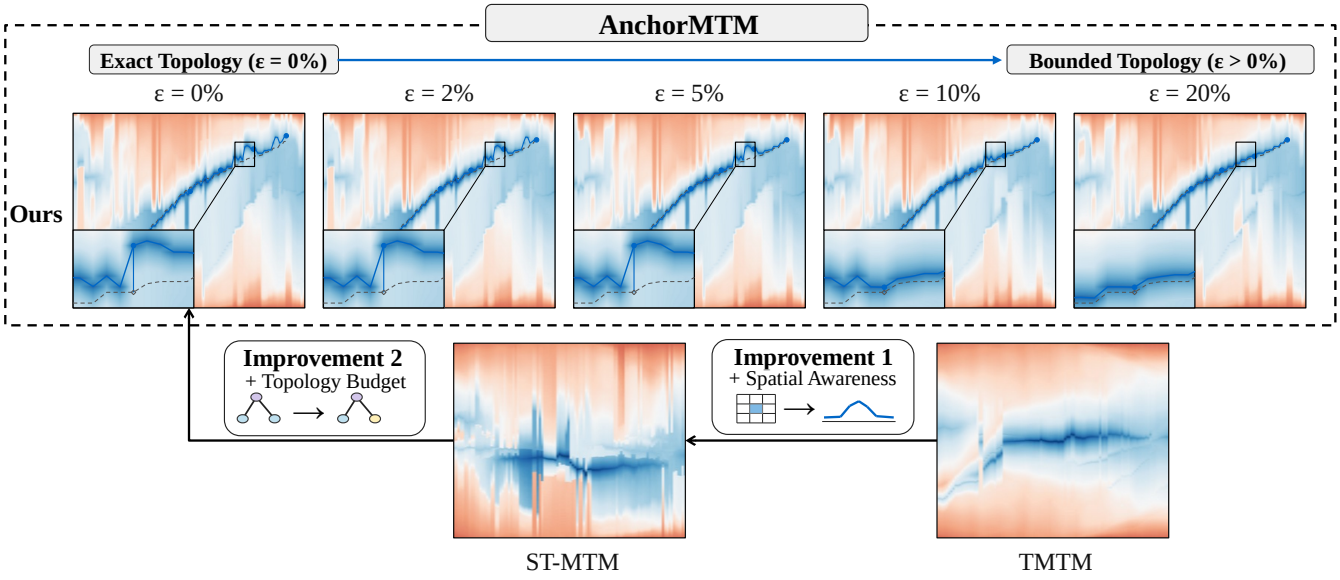


Figure 8: Storms parameter maps and method comparison. AnchorMTM spans exact topology ($\varepsilon = 0$) and bounded topology (2, 5, 10 and 20%), with the original trajectories and common local crops. TMTM and ST-MTM maps use the same scalar fields and augmented trees through local implementations. Spatial awareness and topology budget indicate the respective conceptual improvements. All maps show frames 355–460 with the original blue–orange scale, $[-35, 35]$ hPa. Module connections summarize design differences rather than exact algebraic equivalence.

Table 3: Two groups of sequences. The bound suite reuses the saved trees of an earlier benchmark and one default configuration; the layout suite reconstructs the ST-MTM benchmark and adds an hourly $PM_{2.5}$ sequence and the TMTM volumes.

Suite	Sequence	Field	Frames	Used in
Bound	Storms [25]	ERA5 sea-level pressure (12 h)	118	Q1, Q2
	Wildfires [19]	fire radiative power, daily	30	
	OLR [33]	outgoing longwave radiation	60	
	$PM_{2.5}$ [11]	$PM_{2.5}$ concentration, daily	366	
	Ring	synthetic ring	40	
	Tangaroa [45]	velocity magnitude (3D)	201	
	Cylinder [8]	Q-criterion (3D)	102	
Layout	Gaussian	three moving peaks	64	Q3, Q4
	Ring	synthetic ring	40	
	Wildfires	fire radiative power (32 bins)	32	
	Storms	ERA5 pressure anomaly (1 h)	744	
	$PM_{2.5}$ [29]	$PM_{2.5}$ concentration, hourly	744	
	Tangaroa	velocity magnitude (3D)	201	
	Cylinder	Q-criterion (3D)	102	

Table 4: Volumetric comparison with a fixed x reference. Bold/underline mark best/second distinct values per dataset; ties share emphasis.

Dataset	TMTM			AnchorMTM (ours)			
	SNS↓	TW↑	NPE↓	Mode	SNS↓	TW↑	NPE↓
Cylinder	0.2956	<u>0.8017</u>	0.4769	Exact	<u>0.1226</u>	0.8010	<u>0.1773</u>
				Balanced	**0.0329**	**0.9117**	**0.0458**
Tangaroa	0.5255	0.6701	0.3866	Exact	<u>0.1204</u>	<u>0.8213</u>	<u>0.1285</u>
				Balanced	**0.1124**	**0.8408**	**0.0653**

All values are local measurements, including adapted TMTM. TW uses $k = 1$, with coverage 102/102 and 201/201. Mode applies only to AnchorMTM. Configurations are frozen exploratory results, not a blind test; settings appear in Supplementary Table S4.

every score is local, ST-MTM also keeps a lower SNS and a higher TW than Exact (0.1897 and 0.8224 versus 0.1964 and 0.7884). Against the curves, Balanced improves SNS and TW on Gaussian, Wildfires and Storms, but Hilbert is better on Ring (SNS 0.1252 and TW 0.9048 versus 0.1334 and 0.8571, with TW from one eligible frame), Balanced raises SNS on Wildfires from 0.0426 to 0.0447, and both curves keep a higher TW than the 5% map on $PM_{2.5}$.

Reading motion. Positions are one reading; another asks whether two features that approach or separate in space also do so on the map. On the four planar bound sequences with such events (Storms, Wildfires, OLR and Ring; 56–124 events in which the distance between two tracked features changes by more than 2% of the domain diagonal within two frames), an anchored exact-topology layout misreads 5–32% of the events, TMTM 39–65% and ST-MTM 21–41%, and neither baseline misreads fewer events on any sequence. The task involves relative distances, which ST-MTM fits directly, and relates to the motion term of our placement, so it supports reading motion rather than providing independent evidence.

Volumes. With the tuned configurations and the $+x$ reference of Table 4, Exact lowers NPE at zero merge error from 0.477 to 0.177 on Cylinder and from 0.387 to 0.129 on Tangaroa, and Balanced improves all three metrics over TMTM; Tangaroa's 33.33% budget and geometry-dominated weights limit attribution. Figure 6 shows the positions behind such numbers under the default configuration. TMTM and ST-MTM gather the Cylinder vortices near the middle of the axis (mean worst displacement 47.7% and 51.6%), whereas Exact already reproduces the shedding pattern of the references (9.2%) and a 5% budget removes almost all of the rest (1.2%). Tangaroa's conflicts are deep: Exact lowers 30.6% (TMTM) and 34.2% (ST-MTM) to 11.2%, and the pattern of the references appears near $\varepsilon = 20\%$ (1.9%).

Budgets. Exact has zero measured merge error on every input. Balanced attains at most 0, 0.982, 4.421 and 9.985%, within its budgets, and all runs pass continuous and raster checks without collapsed anchors; Ring needs checked joint integer rounding in 26 Exact and 21 Balanced frames. In the volumetric sweeps, all 1,515 rendered frames were rechecked against the fixed-leaf merge matrices without a violation. The checks cover diagonal extrema,

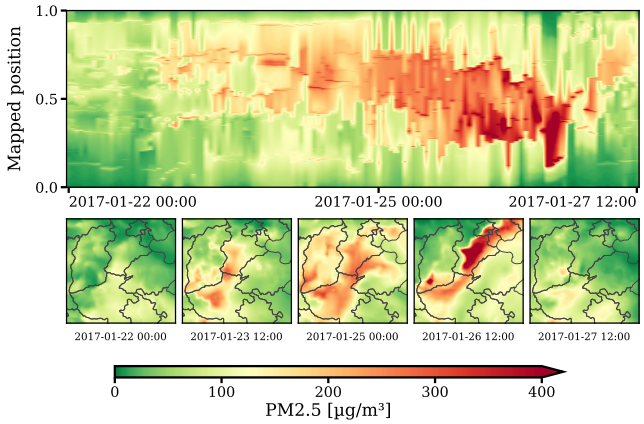


Figure 9: Hourly $PM_{2.5}$, January 22 00:00 to 27 12:00, 2017 (Beijing time): 133 columns of the $\varepsilon = 5\%$ map and five source snapshots; values above 400 $\mu g/m^3$ saturate.

not every augmented internal label.

6.5 Q4: What Does a Budget Buy?

Figure 7 shows the budget frame by frame; what it buys depends on how many frames contain a conflict that it can release. On Storms 35% of the frames use the allowance at $\varepsilon = 10\%$, in a few runs of consecutive frames (Supplementary Fig. S2), so the budget shortens the upper tail: the median per-frame displacement moves only from 1.10% to 0.84%, while the 90th percentile falls from 10.7% to 2.5% and the worst frame from 44.8% to 11.5%. On Wildfires the same 7 of 32 frames use the allowance at every budget, and on Ring the median falls from 3.9% to 1.0% while the last frame stays at 10.0% until the budget reaches 63% (Supplementary Fig. S1). On hourly $PM_{2.5}$, 97% of the frames use the 5% budget and the whole distribution moves: the median falls from 14.5% to 2.1%. So it does in the volumes, where 200 of 201 Tangaroa and 98 of 102 Cylinder frames use a 2% budget: the Cylinder median falls from 6.8% to 1.0% already at 2%, and the Tangaroa median from 11.3% to 1.2% at 20%. A budget limits merge error, not displacement, so a positive budget need not change the map, as on Gaussian.

6.6 Reading the Maps

The upper row of Fig. 8 shows a qualitative Storms archive, prepared separately from Table 2, at budgets from 0 to 20%. The 0, 2 and 5% layouts overlap locally, while 10 and 20% move the selected storm toward its fixed reference, so its track can be read as motion along the reference direction. The lower row shows TMTM and ST-MTM maps recomputed on the same 744 fields and augmented trees, in the same window and colours: TMTM aligns subtrees by temporal overlap and ST-MTM places them by relative distance, so a position on their axes does not correspond to a location of the storm. Both baselines are local implementations, not author binaries; genuine multiway saddles are handled by equal-level separators (TMTM) or an equal-level binary refinement whose leaf merge matrices are checked (ST-MTM).

Figure 9 shows a pollution episode in the hourly $PM_{2.5}$ map at $\varepsilon = 5\%$: it builds from January 23, peaks on January 26, when its strongest features move toward lower reference positions, and clears by January 27. The attained merge error stays within the budget on all 744 frames (at most 4.998%).

Parameters and runtime. One cold Balanced run takes 0.5–63 s on an Apple M4, of which ordering and placement take 0.15–3.3 s; the rest is preprocessing, tree extraction, matching, rasterization and checks. The 744-frame $PM_{2.5}$ run at 5% takes 231 s, 200 s of it in ordering and placement. Supplementary Tables S3 and S4 list all settings, and Supplementary Fig. S8 shows a joint guidance–topology sweep on Ring with source snapshots.

7 Discussion and Limitations

Using the bounds. The bounds split any map's error into a part the tree forces and a part its layout chooses. Order error above ν^* can always be removed without changing the tree, since midpoint ordering attains ν^*; the width-aware and bounded-topology bounds need not be attained, so an excess above them may lie partly in the bound. Because they need only the tree, the projected locations and the widths, the bounds can be computed before a map is drawn, to choose a direction or to decide whether a budget is worth spending, and any new layout of this family can be measured against them.

Position and neighborhoods. A fixed spatial reference and a geometry-oriented embedding answer different questions. AnchorMTM supports reading positions and motion along a chosen direction; ST-MTM's higher reported TW shows the value of preserving local neighborhoods, and neither property guarantees the other. Our results support this task-specific claim, not universal superiority in spatial or temporal fidelity. One axis also carries one coordinate: reading two-dimensional positions would need a second map along another direction, with its own bound.

Limitations. Author-processed fields and their small-sample TW protocol are not fully available, so published SNS and TW cannot support an identical-input significance test, and the two suites differ in preprocessing. The reference and the tracks are computed offline from the whole sequence, and support overlap is a correspondence proxy, not a physical identity. Order search is heuristic, and a continuous certificate does not remove raster rounding, so we check the rendered merge matrices but not every augmented internal label. The motion reading assumes known identities. We have not measured perceptual readability, the accuracy of human motion judgments or generalization to unseen data; these need user studies.

8 Conclusion

Anchoring the axis of a merge tree map to space has an unavoidable positional price. Writing merge tree maps as one layout problem, with TMTM and ST-MTM as exact-topology instances, we bound this price for every instance: the bounds are attained on 606 of 614 planar benchmark frames, and 31–95% of the order error of TMTM and ST-MTM is not forced by the tree. AnchorMTM pays little more than the price: at the same zero merge error it lowers the mean worst displacement by 73–96% on the four ST-MTM benchmark inputs and hourly $PM_{2.5}$, and a declared merge budget, verified on every rendered frame, buys further position where conflicts are shallow, while ST-MTM's published neighborhood preservation remains higher.

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